



Intrinsically flame retardant cardanol-based epoxy monomer for high-performance thermosets

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ABSTRACT

Petroleum-based diglycidyl ether bisphenol A (DGEBA)-type epoxy is one of the most popular thermosetting materials currently, but it suffers from brittleness, flammability and non-renewability. In this study, an intrinsically flame retardant bio-based epoxy monomer (HECarCP) was synthesized from cardanol and cyclophosphazene. Subsequently, HECarCP was cured by 4,4'-diaminodiphenyl methane (DDM) to obtain a thermosetting material. Compared to the cured DGEBA/DDM thermoset, the cured HECarCP/DDM thermoset showed lower initial decomposition temperature; however, the earlier thermal decomposition caused by cyclophosphazene core contributed to char formation in return. As well, the HECarCP/DDM exhibited slightly lower tensile strength but much higher elongation at break than the DGEBA/DDM, overcoming the drawback of brittleness of the DGEBA/DDM. More importantly, the cyclophosphazene core provided intrinsic flame retardancy to the HECarCP/DDM thermoset, which displayed a relatively high LOI value of 33.0% and UL-94 V-0 classification. In cone calorimeter measurement, the peak heat release rate, total heat release and total smoke production of the HECarCP/DDM thermoset was declined by 63%, 24% and 72%, respectively, compared to those of the DGEBA/DDM, owing to the formation of a thermally stable and insulating char layer during combustion. This study paves the way for broadening applications of cardanol in fabrication of high-performance bio-based thermosetting polymers with excellent comprehensive properties.

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1. Introduction

Epoxy resin (EP), as one of the most common thermosetting polymer materials, possesses many virtues including excellent adhesion, high mechanical strength and stiffness, high thermal stability, low dielectric constant and dielectric loss, as well as good dimensional stability. Therefore, it is widely used in high-tech fields such as integrated circuits and composite materials [1–5]. According to the Research Report of Business Communications Company (BCC), the global market sales of epoxy thermosetting materials in 2015 was \$7 billion, and it is expected to reach \$10.2 billion in 2021, with an average annual compound growth rate of about 6.3% [6]. Currently, 90% of the world's epoxy resins are diglycidyl ether bisphenol A (DGEBA)-type epoxy, which are prepared by substitution reaction of epichlorohydrin and bisphenol A (BPA). Although DGEBA-type epoxy has many excellent properties aforementioned, it also suffers from brittleness, flammability and non-renewability. Therefore, in the face of the decreasing petroleum-based resources and the increasing concerns on sustainable de-

velopment, the preparation of high-performance bio-based epoxy thermosets has been becoming a hot issue in the field of bio-based polymer materials.

Until now, there are several reports regarding the preparation of flame retardant bio-based epoxy monomers from fatty acid, eugenol, vanillin, daidzein, itaconic acid, furan, etc [7]. Based on the chemical structures, flame retardant bio-based epoxy monomers can be divided into two categories: aromatic bio-based epoxy monomers and aliphatic ones. For the flame retardant aromatic bio-based epoxy monomers, Dai et al. synthesized a daidzein diglycidyl ether (DGED) and then cured DGED with 4,4'-diaminodiphenylmethane (DDM) to obtain a thermoset [8]. DGED/DDM passed V-0 level in UL-94 vertical combustion test, and its LOI value was as high as 31.6%, while DGEBA/DDM thermoset had a low LOI of 24.5% and no grade in UL-94 vertical combustion test. However, the high content of aromatic structure of daidzein might result in brittleness of this bio-based epoxy thermoset. Wan et al. synthesized eugenol-based bifunctional epoxy monomer (TPEU-EP), and then cured with 3,3'-diaminodiphenylsulfone (33DDS) [9]. The LOI of TPEU-EP/33DDS system was 26.8%, slightly higher than that of DGEBA/33DDS system (23.5%). The peak heat release rate (PHRR) of TPEU-EP/33DDS

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